Stochastic Differential Equations

math-foundations / 33-sde

1 From ODE to SDE

Classical ODEs $\dot{x} = b(x,t)$ describe smooth deterministic dynamics. To model systems perturbed by "continuous" noise, replace the increment $dx = b(x,t)\,dt$ with

$$dX_t \;=\; b(X_t,t)\,dt \;+\; \sigma(X_t,t)\,dW_t,$$

where W_t is a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$. Because Brownian paths have unbounded total variation, this expression is shorthand for the Itô integral equation

$$X_t \;=\; X_0 + \int_0^t b(X_s,s)\,ds + \int_0^t \sigma(X_s,s)\,dW_s.$$

Definition 1 (SDE, drift, diffusion)**.** *A (one-dimensional, time-inhomogeneous)* Itô SDE *is the relation $dX_t = b(X_t,t)\,dt + \sigma(X_t,t)\,dW_t$ with X_0 an $\mathcal{F}_0$-measurable random variable. The map $b : \mathbb{R}\times[0,T]\to\mathbb{R}$ is the* drift coefficient*; $\sigma : \mathbb{R}\times[0,T]\to\mathbb{R}$ is the* diffusion coefficient*. A* strong solution *is an adapted continuous process X_t satisfying the integral equation a.s.*

Remark 1 (Existence/uniqueness)**.** *If b,σ are uniformly Lipschitz in x and of linear growth, then for any $X_0 \in L^2$ independent of W there exists a unique strong solution with $\mathbb{E}\sup_{t\le T}|X_t|^2 < \infty$. The proof is Picard iteration on the Banach space $L^2(\Omega; C([0,T]))$ with the Itô isometry replacing $|\cdot|$ estimates in the stochastic integral.*

2 Discretization: Euler–Maruyama

Definition 2 (Euler–Maruyama scheme)**.** *Fix $N \in \mathbb{N}$, $\Delta t = T/N$, $t_n = n\,\Delta t$. The* Euler–Maruyama *(EM) discretization of $dX_t = b(X_t,t)\,dt + \sigma(X_t,t)\,dW_t$ is*

$$X_{n+1} \;=\; X_n \;+\; b(X_n,t_n)\,\Delta t \;+\; \sigma(X_n,t_n)\sqrt{\Delta t}\,Z_n, \qquad Z_n \overset{iid}{\sim} \mathcal{N}(0,1).$$

Theorem 1 (Strong convergence of Euler–Maruyama)**.** *Let $b,\sigma : \mathbb{R}\times[0,T]\to\mathbb{R}$ be uniformly Lipschitz in x, $\frac{1}{2}$-Hölder in t, and of linear growth. Let X_t be the strong solution of $dX_t = b(X_t,t)\,dt + \sigma(X_t,t)\,dW_t$ on $[0,T]$ with $X_0 \in L^2$, and let $\{X_n\}_{n=0}^N$ be the EM approximation on the same Brownian path with step $\Delta t = T/N$. Then there exists a constant $C = C(T, K, \mathbb{E}|X_0|^2)$ such that*

$$\mathbb{E}\big[\,|X_{t_n} - X_n|\,\big] \;\le\; C\,\sqrt{\Delta t} \qquad \text{for all } 0 \le n \le N.$$

That is, EM has strong order $1/2$*. (Its* weak order *is 1.)*

Heuristic sketch. Fix a step $[t_n, t_{n+1}]$ and write $\Delta W_n := W_{t_{n+1}} - W_{t_n}$, so $\Delta W_n \sim \mathcal{N}(0, \Delta t)$ and $\mathbb{E}[(\Delta W_n)^2] = \Delta t$. Taylor-expand the exact increment of X:

$$X_{t_{n+1}} - X_{t_n} = \underbrace{b(X_{t_n}, t_n)\,\Delta t}_{A_n} + \underbrace{\sigma(X_{t_n}, t_n)\,\Delta W_n}_{B_n} + R_n,$$

where R_n collects higher-order Itô–Taylor corrections. The EM increment keeps exactly $A_n + B_n$, so the local error is R_n. By Itô–Taylor expansion,

$$R_n \;=\; \tfrac{1}{2}\sigma\,\partial_x\sigma\big((\Delta W_n)^2 - \Delta t\big) \;+\; O\big((\Delta t)^{3/2}\big) \;+\; \text{terms of order } \Delta t \cdot \Delta W_n.$$

The Brownian-increment scaling $\Delta W_n \sim \sqrt{\Delta t}$ implies:

- the pathwise (strong) local error has size $\sqrt{\Delta t} \cdot \Delta t = (\Delta t)^{3/2}$ per step from the $\Delta t\,\Delta W_n$ term, summed over $N = T/\Delta t$ steps and combined with Gronwall on the Lipschitz bound, this gives global strong error $\le C\sqrt{\Delta t}$;
- the mean-zero terms $(\Delta W_n)^2 - \Delta t$ and ΔW_n *cancel in expectation*, so for test functions the leading error reduces by a half-power, yielding weak order 1.

A Grönwall argument on $e_n := \mathbb{E}|X_{t_n} - X_n|^2$, using the Lipschitz hypothesis on b, σ and the Itô isometry, then closes the recursion $e_{n+1} \le (1 + C\Delta t)\,e_n + C(\Delta t)^2$, giving $e_n \le C\Delta t$, hence $\mathbb{E}|X_{t_n} - X_n| \le C\sqrt{\Delta t}$. □

Remark 2 (Higher-order schemes)**.** *Adding the Itô–Taylor term* $\frac{1}{2}\sigma\,\partial_x\sigma\,((\Delta W_n)^2 - \Delta t)$ *to the increment gives the* Milstein scheme*, with strong order* 1.

3 Worked example: Ornstein–Uhlenbeck

For $dX_t = -\theta X_t\,dt + \sigma\,dW_t$, the integrating factor $e^{\theta t}$ gives

$$X_t = X_0 e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)}\,dW_s, \quad \operatorname{Var}(X_t) = \frac{\sigma^2}{2\theta}\big(1 - e^{-2\theta t}\big) \xrightarrow{t\to\infty} \frac{\sigma^2}{2\theta}.$$

The companion code verifies the stationary variance empirically by averaging X_T^2 over many EM-simulated paths at large T.